

Seungdong Yoa

CONTACT INFORMATION

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RESEARCH INTERESTS

Agentic AI & Multi-Agent Reasoning
Self-Evolving Agents & Foundation Models
Efficient AI & LLM Quantization
Anomaly Detection & LLM Reliability (Hallucination Detection)

EDUCATION

Korea University, Seoul, Korea

M.S. in Computer Science and Engineering Sep 2019 – Feb 2022
GPA: 4.02 / 4.5 — Advisor: Hyunwoo J. Kim, Ph.D

Korea University, Seoul, Korea

B.S. in Computer Science and Engineering (Major) Mar 2010 – Feb 2017
B.A. in Business Administration (Minor)
The National Scholarship for Science and Engineering (Full Merit-based)

PUBLICATIONS

1. **Seungdong Yoa**, Ye Seul Sim, Suhee Yoon, Sanghyu Yoon, Dongmin Kim, Soonyoung Lee, Junhyun Lee, Bumsoo Kim, “ASQ: Agent-guided Semantic-aware Quantization for Large Language Models.”, *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.
2. **Seungdong Yoa**, Ye Seul Sim, Suhee Yoon, Sanghyu Yoon, Dongmin Kim, Soonyoung Lee, Bumsoo Kim, Junhyun Lee, “Efficient Multi-Agent Reasoning via Confidence-Guided Adaptive Debate.”, *International Conference on Machine Learning (ICML)*, 2026.
3. **Seungdong Yoa**, Sanghyu Yoon, Suhee Yoon, Dongmin Kim, Ye Seul Sim, Junhyun Lee, Woohyung Lim, “From Static Benchmarks to Dynamic Protocol: Agent-Centric Text Anomaly Detection for Evaluating LLM Reasoning.”, *International Conference on Learning Representations (ICLR)*, 2026.
4. **Seungdong Yoa**, Seungjun Lee, Hyesung Cho, Bumsoo Kim, Woohyung Lim, “ImagePiece: Content-aware Re-tokenization for Efficient Image Recognition.”, *AAAI Conference on Artificial Intelligence (AAAI)*, 2025.
5. **Seungdong Yoa**, S. Lee, C. Kim, H. J. Kim, “Self-Supervised Learning for Anomaly Detection With Dynamic Local Augmentation.”, *IEEE Access*, 2021.
6. **Seungdong Yoa**, J. Park, H. J. Kim, “Learning Non-Parametric Surrogate Losses With Correlated Gradients.”, *IEEE Access*, 2021.
7. **Seungdong Yoa**, M. Jeon, Y. Oh, H. J. Kim, “Learning to Balance Local Losses via Meta-Learning.”, *IEEE Access*, 2021.

EMPLOYMENT & EXPERIENCE

LG AI Research, Seoul, Korea
AI Research Scientist

2022 – Present

- **Recursive Self-Quantization (RSQ)**: Exploring failure-driven, training-free behavioral adaptation through self-evolving calibration environments and adaptive precision allocation.

- **Agent-driven Dynamic Workflows (AnoX):** Developed an AI agent platform orchestrating anomaly detection workflows dynamically using the Model Context Protocol (MCP), decoupling agents from tools for zero-code extensibility.
- **LLM Reasoning & Multi-Agent Systems:** Designed ATAD, a multi-agent benchmark protocol for evaluating LLM reasoning in text anomaly detection (**ICLR 2026**).
- **LLM Reliability & Hallucination Detection:** Addressed numerical hallucination in Business Intelligence using agent-based token efficiency and a semantic-aware validation system.
- **Efficient LLM Reasoning (LASE):** Proposed a leader-centric multi-agent debate framework triggering interaction only in non-neutral regimes. Achieved multi-agent-level reasoning accuracy at near single-agent token cost (**ICML 2026**).
- **Semantic-aware LLM Quantization (ASQ):** Engineered an agent-guided INT4 PTQ framework. Preserved domain-critical knowledge, matching or surpassing the zero-shot accuracy of dense FP16 models with zero inference overhead (**NeurIPS 2026**).
- **Efficient Vision Models:** Authored “ImagePiece,” a re-tokenization framework achieving SOTA efficiency in ViTs (1.5× faster, +0.39% accuracy), topping lightweight leaderboards (**AAAI 2025**).
- **Industrial AI Optimization:** Deployed the ‘Hero LensAA’ project by applying Tab-Transformer and domain-specific regression adjustments, reducing overkill from 99% to near 0% on test sets. Awarded the **2024 LG Award**.

Hyundai Motor Group AIRS Company, Seoul, Korea July 2021 – Sep 2021
AI Researcher Intern

- Developed Anomaly Detection algorithms for Smart Factory quality control via self-supervised learning.

Korea University (M.S.), Seoul, Korea Sep 2019 – Feb 2022
Graduate Research Assistant

- Led a joint AI project with Samsung Electronics, developing real-time human-object interaction and anomaly detection algorithms for industrial drone monitoring systems.

Shinsegae E-commerce SSG.COM, Seoul, Korea Feb 2017 – Jun 2019
Software Engineer

- Engineered an OAuth 2.0 server and a scalable integration platform for AI speakers (Google Home, Naver Clova), and developed a customer service chatbot (CS Talk).

ENTREPRENEURIAL **InfiniTensor Inc.**, Seoul, Korea 2024 – Present
 EXPERIENCE *Founder & CEO*

- **AgentMon:** Built and launched an autonomous marketing agent that receives commands and schedules through Discord and persistently executes content-posting workflows on X (Twitter), combining agent orchestration, tool use, and scheduled execution.
- **EffectGen / EffectGen HomeLiving:** Built generative-AI products for e-commerce businesses that transform product-only images into on-model fashion imagery and contextual home-living scenes for commercial product pages.

HONORS AND AWARDS	LG Award (2024): The most prestigious honor across all LG Group affiliates. Recognized for major contributions and successful commercial deployment in the Hero LensAA project. AI Grand Challenge 2nd Place (Jan 2020): Awarded \$0.6M USD research grant for AI problem-solving excellence. National Scholarship for Science and Engineering (Full Merit-based, Top-tier academic performance), Korea	
EXTRACURRICULAR ACTIVITIES	Future Entrepreneurs' Society (FES) Sep 2014 – Dec 2015 <i>Leading Member / HR Manager</i> - Designed manufacturing startup business models and led business strategy seminars in collaboration with top tech companies (Google Korea, Kakao). Student Council Mar 2010 – Dec 2011 <i>General / Planning Manager</i> - Organized large-scale student activities and managed the council budget.	
PATENT	Wet type vacuum cleaner - Registration number/date: 1010156040000 / 2011.02.10 Wet type air cleaner - Registration number/date: 1009762030000 / 2010.08.10	
ADDITIONAL INFORMATION	Skills: Python, PyTorch, SQL, Next.js, Java, Spring Framework, L^AT_EX. Clubs & Communities: FES Business Strategy Club, Korea Student Aid Foundation (KOSAF), KUBA, Student Council, Soccer Club.	